A Comprehensive Review of Machine Learning and Deep Learning Applications

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Abstract. Data Science is a rapidly growing field that attracts significant attention from researchers due to its potential to extract meaningful insights from vast amounts of data. At the core of Data Science lie Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL), which act as its pillars. In this article, we provide a comprehensive overview of ML and DL, exploring their fundamental concepts, techniques, and practical applications. The discussion highlights how ML enables systems to learn from data, identify patterns, and make accurate predictions, while DL, a specialized subset of ML, utilizes multi-layered neural networks to automatically extract complex features from large and unstructured datasets, such as images, audio, and text. In now days, deep learning has become a prominent and emerging research area in computer vision .Furthermore; the article examines the diverse applications of these technologies across different domains, sub domains and newer areas of applications of deep learning.

Keywords: Machine Learning, Deep Learning, Artificial Intelligence.

1 Introduction

Artificial Intelligence (AI) is the art of enabling machines to perform tasks that require human thinking abilities, such as learning, decision making, and problem solving [1]. Learning is the main hallmark of human intelligence and the basic means to obtain knowledge. Machine learning is the fundamental way to make the computer intelligent [2].Therefore, Machine Learning (ML) is a subfield of AI, that eliminates the need for explicit programming by allowing systems to automatically identify patterns in data and gradually enhance performance. On the other hand, Deep learning (DL) is a subset of ML. It uses neural networks with multiple layers (hence “deep”) to model complex patterns in large datasets. DL has enabled many practical applications of ML and by extension the overall field of AI [3]. The relationship between AI, ML, and DL, showing that ML is a subset of AI and DL is a specialized subset of ML for complex tasks, is illustrated in Fig. 1.

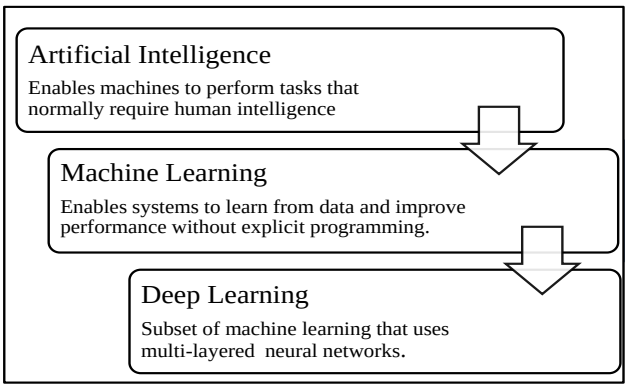

Fig. 1. Relationship between AI, ML &DL.

This article, highlight the techniques and applications of DL and ML .Section 2 provides an overview of Machine Learning, followed by Section 3, which discusses its key applications. Section 4 presents an overview of Deep Learning, while Section 5 highlights its major applications. In the end, we can explore anyone of the newer areas of applications of deep learning, which will yield better results and will add on to the ongoing research in this field.

2 Overview of Machine Learning

Machine learning is based on algorithms that can learn from data without relying on rules-based programming [4]. The central premise of ML is that if you optimize a model's performance on a dataset of tasks that adequately resemble the real-world problems it will be used for—through a process called model training—the model can make accurate predictions on the new data it sees in its ultimate use case. So, any progress of human in the field of machine learning, will enhance the capability of computers, thus have an impact on human society [2]. ML can be generally divided to different categories: supervised learning, unsupervised learning, semi-supervised Learning, Self-Supervised Learning and reinforcement learning, listed in Table 1.

Table 1. Categories of Machine Learning

Category	**Description**
Supervised Learning	Supervised learning uses labeled data to make predictions.
Unsupervised Learning	Uses unlabeled data to discover hidden patterns
Semi-Supervised Learning	Methods use both labeled data and unlabeled data.
Self-Supervised Learning	Supervisory signal is obtained directly from unlabeled data.
Reinforcement Learning	Focuses on how agents make decisions through trial and error.

Semi-supervised learning, Transductive Inference, On-line learning and Active Learning are less widely use [5].

In supervised learning, function that converts inputs into desired outputs is produced by the different algorithms. Modern deep learning models as well as a variety of conventional ML models that are still commonly used in many industries have been generated by supervised machine learning algorithms. The classification problem is a common way to formulate the supervised learning task: the learner must examine multiple input-output examples of a function that maps a vector into one of several classes in order to learn (approximate the behavior of) the function [6]. Modern machine learning term employs "supervision" or "supervisory signals" to refer essentially to any source of fundamental information in order to allow a more flexible concept of supervised learning. Supervised learning algorithms encompass a variety of techniques, including, linear regression, logistic regression, artificial neural networks, and traditional classification algorithms include support vector machines.

— Linear Regression (LR): LR predict continues values such as price, duration, temperature and size. It shows a linear relationship between one or more independent variables and a numeric dependent variable outcome [7].

$$y = \alpha + \beta x \tag{1}$$

Here, α stands for the intercept, or the value of y that should be expected when x = 0, and β for the slope, which shows how y changes for every unit change in x. The goal of linear regression is to determine the optimal values of α and β that minimize the discrepancy between the actual and predicted values of y, typically by reducing the sum of squared errors.

— Logistic Regression (LR): Joseph Berkson was the primary developer and pioneer of the logistic regression as a general statistical model, coming up with the term "Logit" in Berkson (1944).LR, predicts the probability of binary (two-outcome) classification problems or categorical event occurring, fitting a curve with a S shape rather than a straight line to the data. It uses *sigmoid* function to convert inputs into a probability value between 0 and 1.The parameters of a logistic regression are most commonly estimated by maximum-like hood estimation (MLE). MLE plays a similarly basic role for binary or categorical responses as linear regression by ordinary least squares (OLS) plays for scalar responses. The logic function is utilized in the modeling process. The Logit transformation (log-odds form) is:

$$logit\ (P) = ln\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n \tag{2}$$

The left side of the equation $ln\left(\frac{P}{1-P}\right)$ represents the log-odds of an event, where P the probability of success and $1-P$ is the probability of failure. This log transformation makes the odds linear and easier to model. The right side, $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n$ is a linear function of predictors, where, β_0 is the intercept and each β_n shows how much the log-odds change with a one-unit increase in the corresponding predictor.

— Artificial Neural Network or Neural Net (ANN or NN): The processing units that make up an artificial neural network are known as neurons. A neuron network which is inspired by the structure and functions of biological neural networks. ANN are use for various task, including predictive modeling, adaptive control, and solving problems in AI. A neuron consists of inputs (dendrites), and one output (synapse via axon) [8]. As seen in Fig. 2.

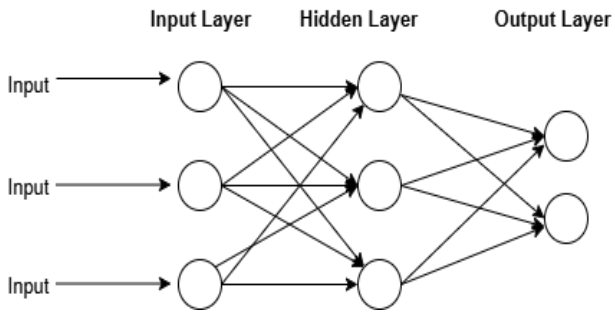

Fig. 2. : Schematic of an artificial neural network

Neural networks include a combination of several layers, including input, output, and hidden layer. Input Layer receives the raw data. Hidden layer process the information from the input layer. Each artificial neuron in these layers receives input signals from previous neurons. Inside each neuron, a non-linear function called an activation function is applied to the weighted sum of inputs, determining the neuron's output signal and the final output layer generate the output.

— Support Vector Machines (SVMs): The major machine learning achievement was SVM, proposed by Vapnik and Cortes in 1995 with solid hypothetical standing and exact outcomes [9]. It works by finding an optimal hyper plane (line or plane) to use in order to maximize the margin between data points of various classes. SVMs can efficiently perform non-linear classification using the kernel trick. SVMs are popular because they can be theoretically analyzed and can be used for a wide range of applications, including structured prediction issues.

In unsupervised learning, models are trained on unlabeled data to uncover hidden patterns, structures, or groupings without predefined output categories. Unsupervised learning algorithms encompass a variety of algorithms, including, but not limited to K-Means Clustering and Dimensionality Reduction Algorithms.

— K-Means Clustering: The term "*k*-means" was first used by James MacQueen in 1967, though the idea goes back to Hugo Steinhaus in 1956.K-unique clusters are automatically formed by this algorithm. It is a type of unsupervised learning where variables in the data are grouped together based on relationships among them [10]. K-means clustering algorithm is an example of an exclusive or “hard” clustering method. In data science, this kind of cluster analysis is frequently used for visual compression, market segmentation, document clustering, and image

segmentation. Due to its simplicity, effectiveness, and efficiency, the k-means algorithm is a frequently used technique in cluster analysis.

— Dimensionality Reduction Algorithms: such as the Principle Component Analysis Algorithm (PCA), that aims to minimize projection error by minimizing the distance between each feature and a specific projection line [11].

A sort of machine learning called reinforcement learning (RL) teaches an agent to make decisions by interacting with its surroundings. The agent tries various actions and is rewarded or penalized according to the results, rather than being given the right answers. It gradually discovers which behaviors maximize its overall reward. An example of a reinforcement learning model is the Markov Decision Process (MDP) [5].

In Self-supervised learning, model is trained using data that does not have any labels or answers provided. Instead of needing people to label the data, the model finds patterns and creates its own labels from the data automatically. This enables the model to teach itself from the data and acquire valuable knowledge. When there is a large amount of data but just a tiny portion of it is labeled, or when labeling the data would require a significant amount of time and effort, SSL is very helpful.

In semi-Supervised Learning, hybrid machine learning approach which uses both supervised and unsupervised learning. It uses a small amount of labeled data combined with a large amount of unlabelled data to train models. The goal is to learn a function that accurately predicts outputs based on inputs, similar to supervised learning, but with much less labeled data. Semi-supervised learning is particularly valuable when acquiring labelled data is expensive or time-consuming, yet unlabelled data is plentiful and easy to collect.

The machine learning algorithms which are widely in use are Decision Trees (DT) [12],Boosting [13], Naïve Bayes (NB) [11],k-Nearest Neighbor (k-NN),and many more.

3 Machine Learning Applications

The various application domains include computer vision, natural language processing, healthcare, and finance.

— Computer Vision (CV): The journey of CV started in 1960, where father of computer vision Larry Roberts proposed 3D geometrical information extraction from 2D perspective polyhedral concepts in his Ph.D. thesis at MIT.

— Natural Language Processing (NLP): efficiently analyzes and interprets vast amounts of textual and speech data by combining machine learning and computational linguistics. When integrated with structured project planning principles, NLP and ML tools can improve surgical workflows by providing accurate time predictions, simplifying documentation, and detecting potential delays or inefficiencies [14].

— Healthcare: ML techniques are widely used in the healthcare industry for a variety of purposes, including drug discovery [15], disease prediction, medical image analysis, and many more. These methods help with early disease diagnosis, en-

hance the precision of medical imaging interpretations, allow for individualized treatment regimens based on patient information, and support predictive analytics for patient outcomes.

— Finance: ML in finance sits at the intersection of a number of emergent and established disciplines including pattern recognition, financial econometrics, statistical computing, probabilistic programming, and dynamic programming [16].

4 Overview of Deep Learning

Hinton and colleagues suggested the concept of DL in the year 2006.DL is a subset of ML. It is a neural network with a large number of layers and parameters. Most deep learning methods use neural network architectures. Therefore it is also referred to as deep neural networks [9]. DL is about automatically learning multiple levels of representations of the underlying distribution of the data to be modeled [17]. DL models can directly map inputs to outputs, reducing the need for intermediate processing steps.

— Convolutional Neural Network (CNNs): CNN was firstly introduced by Kunihiko Fukushima [18]. The class of deep learning algorithms most commonly used for spatial pattern analysis is Convolutional neural networks (CNNs or ConvNets). CNNs are designed to learn the spatial features, e.g. edges, corners, textures, or more abstract shapes, that best describe the target class or quantity [19].
— Recurrent Neural Networks (RNNs): RNN's provide a very elegant way of dealing with (time) sequential data that embodies correlations between data points that are close in the sequence [20]. RNN's are powerful learning models that achieve state-of-the-art results in a wide range of supervised and unsupervised machine learning tasks. They are suited especially well for machine perception tasks, where the raw underlying features are not individually interpretable [21].
— Bidirectional Recurrent Neural Networks (BRNNs): Mike Schuster and Kuldip Paliwal first presented the Bidirectional Recurrent Neural Networks (BRNN) in 1997. It trains the network in both positive and negative time directions at the same time [9]. BRNN's that can be trained using all available input information in the past and future of a specific time frame [20].
— Generative Adversarial Networks (GANs): GANs are a powerful class of generative models introduced in 2014 by Good fellow et al [22].Two neural networks are used, both of which were trained concurrently in a competitive environment: Generator: Produces computer-generated data that resembles real data, such as text, audio, and images. Discriminator: check the data whether it is real or fake.

5 Deep Learning Applications

Deep learning has transformed multiple domains by enabling systems to process large, complex, and unstructured datasets with high accuracy. Key applications include:

- Computer Vision: In deep learning, it is more than just identifying objects in images. Additionally, it is able to detect tiny details that humans might overlook, such as slight facial expressions or tiny variations in medical scans. It converts unprocessed images into information that can be used to make decisions in the real world by learning directly from pixels. CV enables systems to extract meaningful information from digital images allowing for detection, identification and automation through integrating inputs from the physical world [23].
- Speech Recognition: With deep learning, computers can comprehend spoken words and convert them into text because of to speech recognition. It learns natural speech patterns, accents, and tones, which makes transcription tools and voice assistants more accurate and human-like than older systems.
- Customer Relationship Management (CRM): Typically used for tasks like sentiment analysis, churn prediction, customer segmentation, and personalized recommendations, it helps businesses in increasing engagement and improving decision-making .
- Prediction: Using neural networks to model complex structures in data and predict future events is known as prediction in deep learning. Recurrent neural networks (RNNs) and long short-term memory (LSTM) networks are two examples of DL models that work especially well with time-series data, customer behavior, and trend analysis. This allows for precise and data-driven decision-making. A prediction domain topic such as Financial Signal Representation and Trading can be very well be implemented using Direct Deep Reinforcement Learning technique in this paper [24].

Applications of deep learning are to be explored in newer areas such as Graphics Processing Unit (GPU) hardware, Natural Language Processing, Computer Vision, Smart Cities, Cultural Heritage, Blood Donation [25],Robotic system and Animal Sound Recognition. The primary reasons why deep learning is well-suited for emerging areas of application include its reliance on large datasets, the use of GPU hardware, and its approach to feature engineering. The term data dependency refers to the ability of DL algorithms to perform effectively when trained on massive amounts of data. GPU is a high-performance computing device that accelerates the training of DL models. Unlike traditional ML, DL can automatically extract high-level features from raw data, a process known as *feature engineering*. Due to these capabilities, DL is expected to find applications in many more areas in the coming years.

6 Conclusion

This article, presented an overview of ML and DL explaining their fundamental concepts, techniques, and main applications. ML, as a branch of AI, helps machines learn from data and improve performance without explicit programming, while DL, as an advanced form of ML, uses multi-layered neural networks to handle complex tasks and unstructured data. Both methods are widely applied in areas such as healthcare, finance, customer management, and prediction systems and they continue to play a

major role in the progress of AI. In the future, exploring new application areas of DL will lead to better results and contribute to ongoing research and innovation in this field.

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